

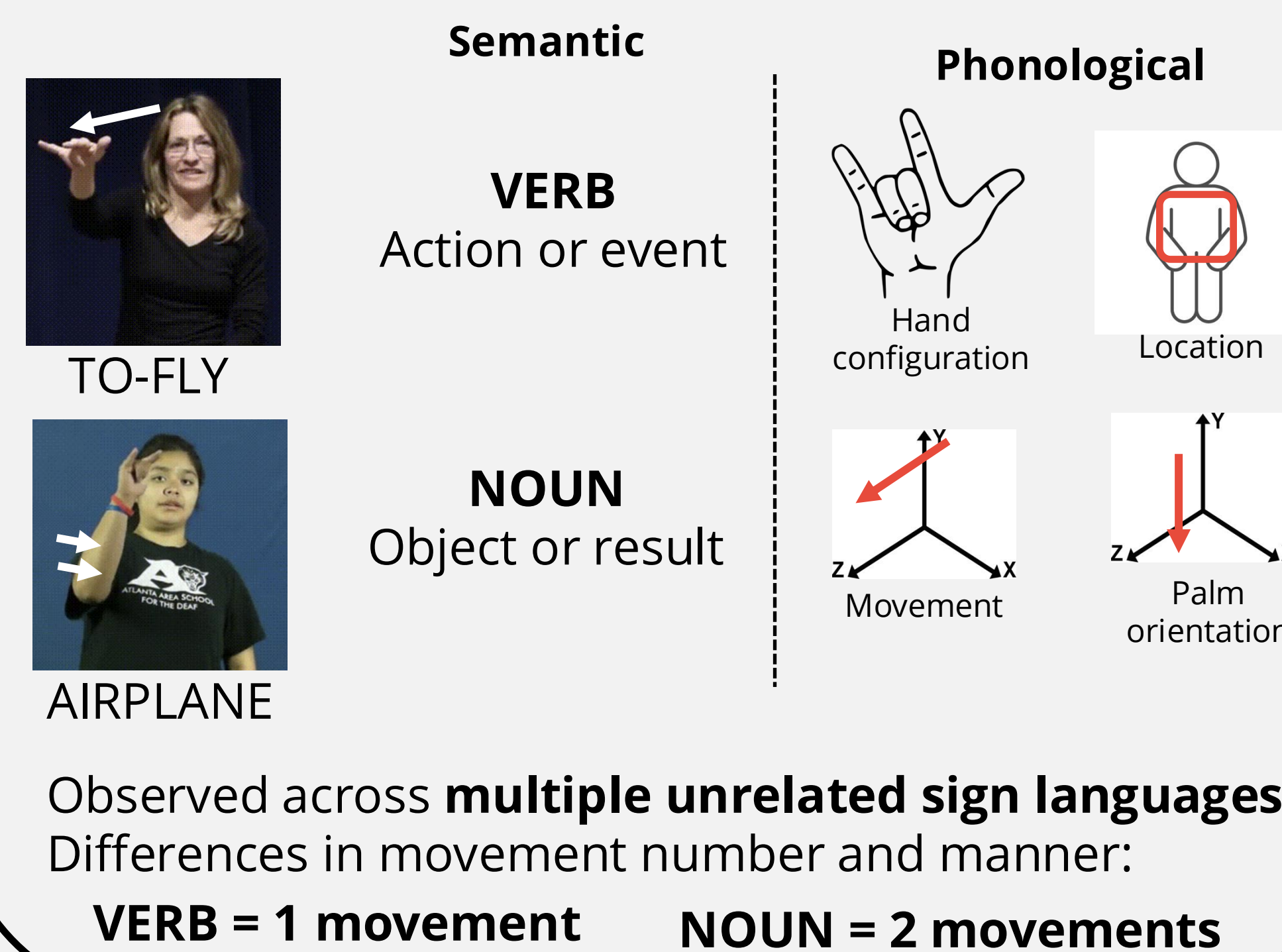
The constructionalization of movement repetition: Explaining noun-verb pairs in sign languages

Tory Sampson^{1, 2}

¹Linguistics, Harvard University, ²Boston University

What are noun-verb pairs?

Two signs that share **semantic** and **phonological** features⁹:

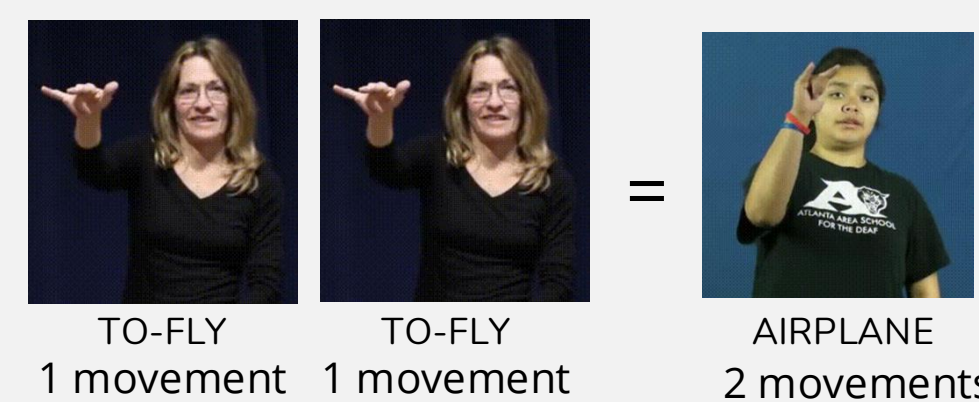


Current analyses

Nominalizing reduplication^{1,9}:
movement repetition as derivational morpheme

Inconsistencies:

- Semi-productive
- Unpredictable semantics
- Not robust in morphosyntactic production
- Omits many other similar sign pairs



Iconicity:

movement repetition: cognitive construal of nouns¹⁰

VERB:

Process distributed across
space & time

NOUN:

Thing in a region of
conceptual space

Inconsistencies:

Hearing non-signers do not perceive movement repetition as **objects**, but as **actions** instead⁷

Constructionalization

As a framework:

Emergence of a derivational morpheme^{2,4}

- Not lexicalization nor grammaticalization

Stages of constructionalization

- Phrasal similes & compounds: **productive**
- New word formation: **productive**
- Loss of productivity** → semantic niches

RQ: How do noun-verb pairs emerge,
and why do they look similar
across sign languages?

Pluractionality: plurality of situations (*time, space, participants*)⁶

IX-1 [WRITE WRITE WRITE]_{pluractional}
'I write a lot'



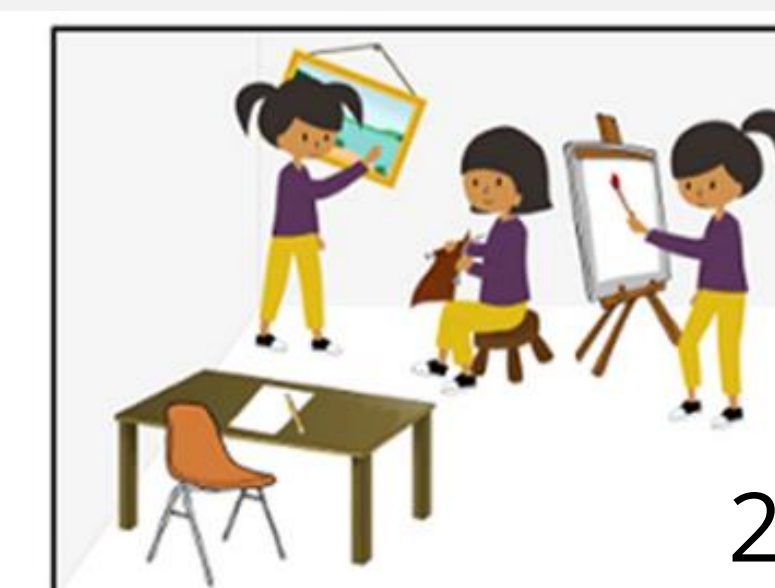
Iconic isomorphism: 1:1 mapping of form & meaning

- Multiple movements = multiple actions
- Iconic resource available to early signers & non-signers
 - Explains why we see noun-verb pairs in many sign languages

Recursive structures: use verbs to index entities

As early as first-generation signers in Nicaraguan SL (LSN)⁵

- Explains why we see noun-verb pairs in many sign languages



1 GIRL DRAW,
GIRL KNIT,
GIRL PAINT

2 GIRL **DRAW** REMOVE PICTURE

Figure 1. Experimental stimuli used in Kocab et al. (2022)

Deverbal anaphor in LSN narratives

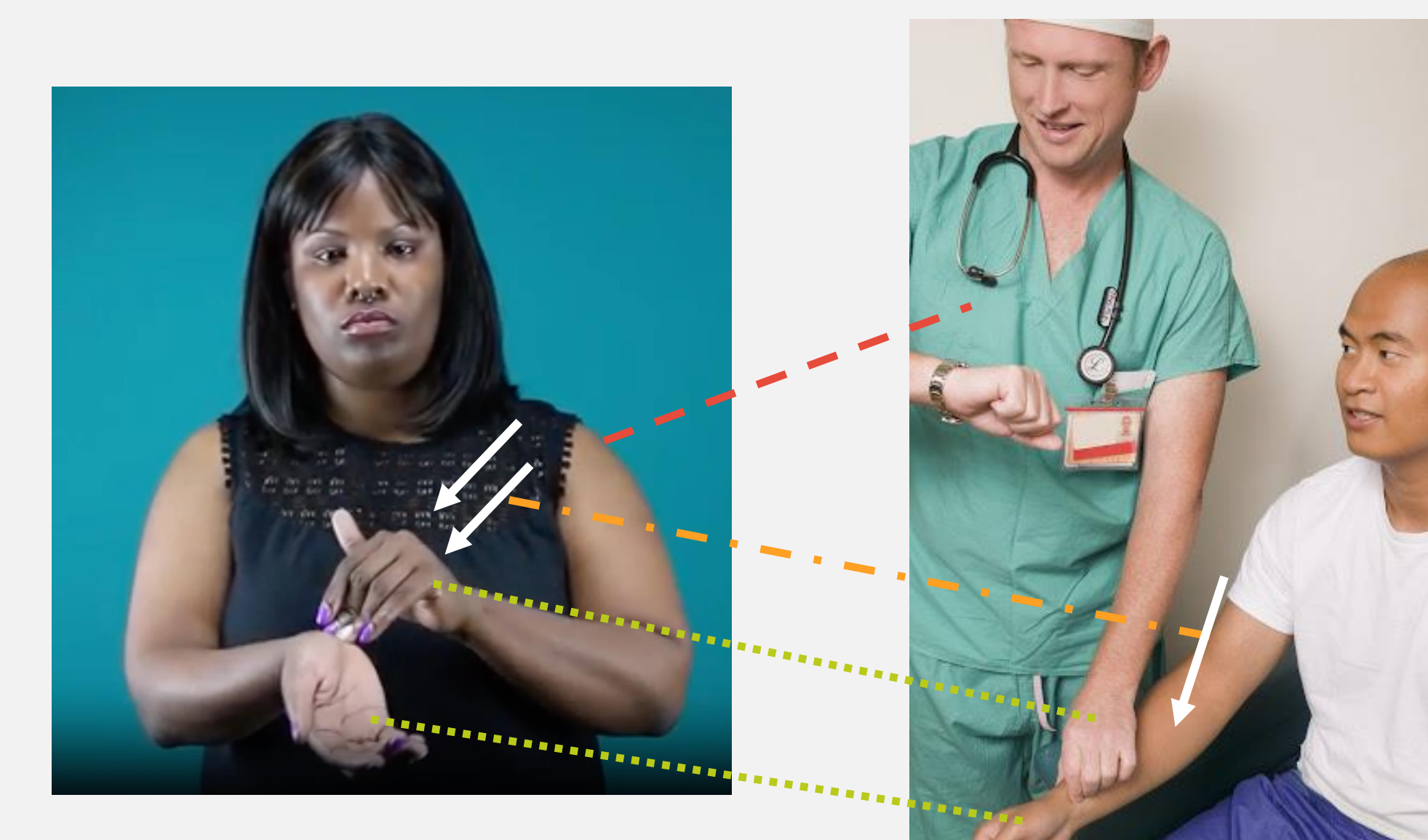
Verbs used to refer to entities doing previously mentioned action

MAN **COLLECT**.pl, BIRD LAUGH, [**COLLECT**]_N LOOK-UP

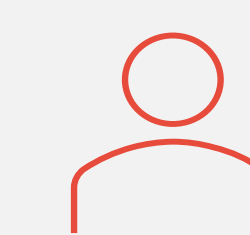
The man was collecting; a bird laughed, and the collector looked up

Hypothesis: Noun signs are reduced forms of pluractional verbs in recursive structures which index the noun

Indexing specific nominal meanings



Sign form component



Signer

'the person who [verb]s repeatedly'



Hands

'the object that is used repeatedly with [verb]'



Movement

(through space & time)

'the activity that results from repeated [verb]'

Noun type

Agentive

'the person who [verb]s repeatedly'

Object

'the object that is used repeatedly with [verb]'

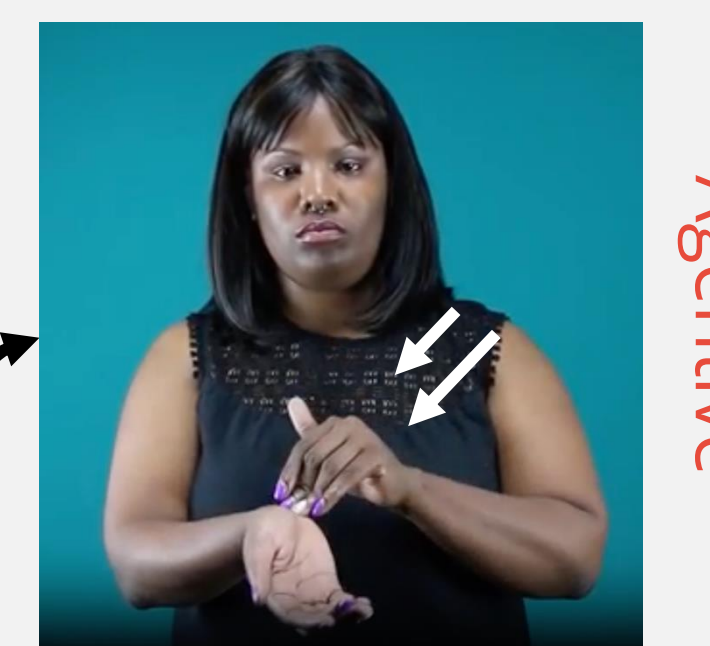
Activity (gerund)

'the activity that results from repeated [verb]'

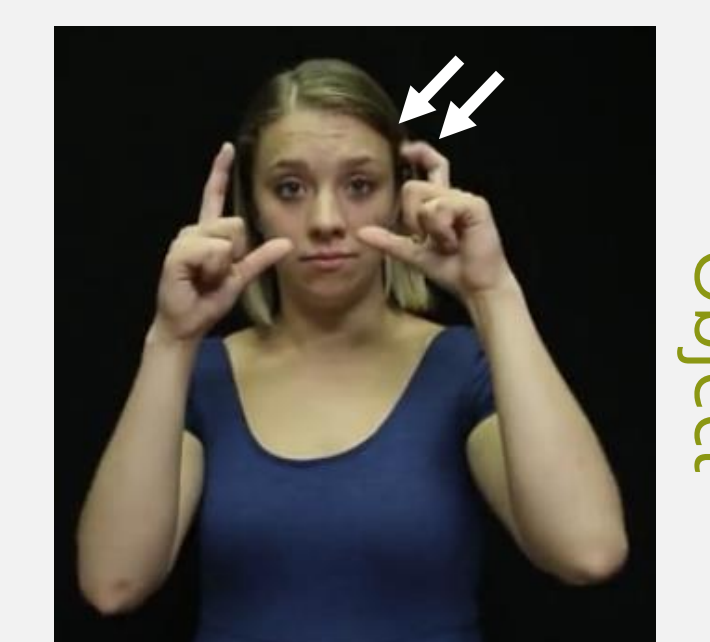
Possible meanings

'a doctor'
'a pulse'
'taking a pulse'

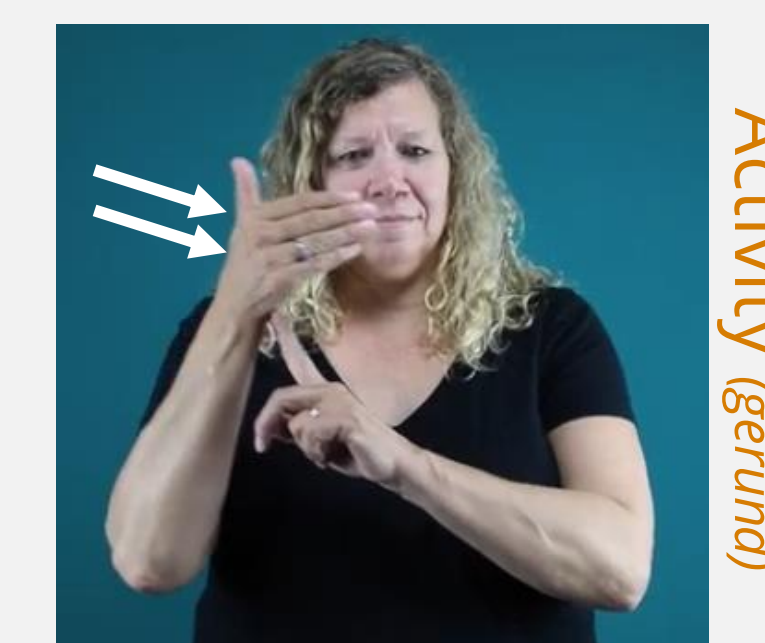
Semantic niches



DOCTOR
(Not 'pulse' or 'taking pulses')



CAMERA
(Not 'photographer' or 'photography')



SAILING
(Not 'sailboat' or 'sailor')

Conventionalization³

Productive use of movement repetition > Loss of productivity

Conclusion:

Movement repetition in noun signs emerged from two combined mechanisms: **pluractionality (iconicity)** and **recursivity (structure)**

And that's why they look similar across unrelated sign languages!

References

- ¹Abner, N. (2017). What You See Is What You Get: Surface Transparency and Ambiguity of Nominalizing Reduplication in American Sign Language. *Syntax*, 20(4), 317–352.
- ²Booij, G. (2010). Construction Morphology. *Language and Linguistics Compass*, 4(7), 543–555.
- ³Bybee, J. (2010). *Language, Usage and Cognition*. Cambridge University Press.
- ⁴Hüning, M., & Booij, G. (2014). From compounding to derivation: The emergence of derivational affixes through "constructionalization." *Folia Linguistica*, 48(2).
- ⁵Kocab, A., Senghas, A., Coppola, M., & Snedeker, J. (2023). Potentially recursive structures emerge quickly when a new language community forms. *Cognition*, 232, 105261.
- ⁶Mattola, S. (2019). *Typology of Pluractional Constructions in the Languages of the World*. John Benjamins.
- ⁷Peters, J. E., & Emmorey, K. (2022). The iconic motivation for the morphophonological distinction between noun-verb pairs in American Sign Language does not reflect common human construals of objects and actions. *Language and Cognition*, 14(4), 622–644.
- ⁸Senghas, A. (1995). *Children's Contribution to the Birth of Nicaraguan Sign Language* [Doctoral]. Massachusetts Institute of Technology.
- ⁹Supalla, T., & Newport, E. (1978). How Many Seats in a Chair? The Derivation of Nouns and Verbs in American Sign Language. In Siple, Patricia (Ed.), *Understanding language through sign language research* (pp. 91–132). Academic Press.
- ¹⁰Wilcox, S. (2004). Cognitive iconicity: Conceptual spaces, meaning, and gesture in signed languages. *Cognitive Linguistics*, 15(2), 119–147.
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